

Accounting Questions

Financial Statements & Accrual Concepts

What is the primary purpose of US GAAP?

In the US, the Securities and Exchange Commission ("SEC") authorizes the Financial Accounting Standards Board ("FASB") to determine the set of accounting rules followed by publicly traded companies.

Under FASB, financial statements are required to be prepared in accordance with US Generally Accepted Accounting Principles ("US GAAP").

Through the standardization of financial reporting and ensuring all financials are presented on a fair, consistent basis – the interests of investors and lenders are protected.



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Basic Accounting

What are the main sections of a 10-K?

In a 10-K, you'll find the three core financial statements, which are the income statement, cash flow statement, and balance sheet. There'll also be a statement of shareholders' equity, a statement of comprehensive income, and supplementary data and disclosures to accompany the financials.

Main Sections of a 10-K	
Business Overview	Overview of the company's business divisions, strategy, product or service offerings, seasonality, geographical footprint, and key risks.
Management's Discussion & Analysis ("MD&A")	Commentary and summarized analysis of the company's fiscal year result from the perspective of the management team.
Financial Statements	The "Core 3": Income Statement, Balance Sheet, Cash Flow Statement The "Other 2": Statement of Comprehensive Income, Statement of Shareholders' Equity
Notes	Supplementary disclosures to the financial statements that provide more details about a company's recent financial performance.

What is the difference between the 10-K and 10-Q?

- **10-K:** A 10-K is the annual report required to be filed with the SEC for any public company in the U.S. The report is comprehensive and includes a full overview of the business operations, commentary on recent performance by management, risk factors, disclosures on changes in accounting policies – and most importantly, the three core financial statements with supplementary data.
- **10-Q:** A 10-Q refers to the quarterly report required to be filed with the SEC. Compared to the 10-K, this report is far more condensed in length and depth, with the focus being on the quarterly financials with brief sections for MD&A and supplementary disclosures.
- **Additional Differences:** A few more differences are 10-Ks are required to be audited by an independent accounting firm, but 10-Qs are only reviewed by CPAs and left unaudited. 10-Ks must also be filed ~60-90 days after the fiscal year ends, whereas 10-Qs must be submitted ~40-45 days after the quarter ends.

Walk me through the three financial statements.

- 1. Income Statement ("IS"):** The income statement shows a company's profitability over a specified period, typically quarterly and annually. The beginning line item is revenue and upon deducting various costs and expenses, the ending line item is net income.
- 2. Balance Sheet ("BS"):** The balance sheet is a snapshot of a company's resources (assets) and sources of funding (liabilities and shareholders' equity) at a specific point in time, such as the end of a quarter or fiscal year.
- 3. Cash Flow Statement ("CFS"):** Under the indirect approach, the starting line item is net income, which will be adjusted for non-cash items such as D&A and changes in working capital to arrive at cash from operations. Cash from investing and financing activities are then added to cash from operations to arrive at the net change in cash, which represents the actual cash inflows/(outflows) in a given period.



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The 3 Financial Statements

Walk me through the income statement.

The income statement shows a company's accrual-based profitability over a specified time period and facilitates the analysis of its historical growth and operational performance. The table below lists the major income and expense components of the income statement:

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The Income Statement

Income Statement Typical Components	
Net Revenue (or Sales)	The income statement begins with revenue (often called the "top line"), which represents the total value of all sales of goods and delivery of services throughout a specified period.
Less: Cost of Goods Sold	COGS represents the costs directly tied to producing revenue, such as the costs of materials and direct labor.
Gross Profit	Revenues – Cost of Goods Sold = Gross Profit
Less: Selling, General & Administrative ("SG&A")	Operating expenses that are not directly associated with the good or service being sold (e.g., payroll, wages, overhead, advertising, and marketing).
Less: Research & Development ("R&D")	R&D refers to developing new products or procedures to improve their existing product/service offering mix.
EBITDA	Gross Profit – SG&A – R&D = EBITDA EBITDA stands for: <u>E</u> arnings <u>B</u> efore <u>I</u> nterest, <u>T</u> axes, <u>D</u> epreciation & <u>A</u> mortization.
Less: Depreciation & Amortization ("D&A")	D&A is a non-cash expense that estimates the annual reduction in the value of fixed and intangible assets
Operating Income ("EBIT")	EBITDA – D&A = Operating Income (or EBIT) EBIT stands for: <u>E</u> arnings <u>B</u> efore <u>I</u> nterest and <u>T</u> axes.
Less: Interest Expense, net	Interest expense from debt, net of interest income generated from investments.
Pre-Tax Income ("EBT")	EBIT – Interest Expense, net = Pre-Tax Income (or "Earnings Before Tax")
Less: Tax Expense	Tax liability recorded by a company for book purposes.
Net Income	EBT – Tax Expense = Net Income (referred to as the "bottom line")

Apple Inc. | Historical Income Statement (Snapshot from [Financial Statement Modeling Course](#))

Apple Income Statement	2016A	2017A	2018A	2019A
(\$ in millions)	9/30/16	9/30/17	9/29/18	12/31/19
Revenue	\$215,639	\$229,234	\$265,595	\$260,174
Less: Cost of Revenue	(131,376)	(141,048)	(163,756)	(161,782)
Gross Profit	\$84,263	\$88,186	\$101,839	\$98,392
Less: Research & Development (R&D)	(10,045)	(11,581)	(14,236)	(16,217)
Less: Selling, General & Administrative (SG&A)	(14,194)	(15,261)	(16,705)	(18,245)
Operating Income (EBIT)	\$60,024	\$61,344	\$70,898	\$63,930
Plus: Interest Income	3,999	5,201	5,686	4,961
Less: Interest Expense	(1,456)	(2,323)	(3,240)	(3,576)
Less: Other Expense, net	(1,195)	(133)	(441)	422
Pre-Tax Income (EBT)	\$61,372	\$64,089	\$72,903	\$65,737
Less: Taxes	(15,685)	(15,738)	(13,372)	(10,481)
Net Income	\$45,687	\$48,351	\$59,531	\$55,256

Walk me through the balance sheet.

The balance sheet shows a company's assets, liabilities, and equity sections at a specific point in time. The fundamental accounting equation is: Assets = Liabilities + Shareholders' Equity. The assets belonging to a company must have been funded somehow, so assets will always be equal to the sum of liabilities and equity.



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Balance Sheet

- **Assets Section:** Assets are organized in the order of liquidity, with "Current Assets" being assets that can be converted into cash within a year, such as cash itself, along with marketable securities, accounts receivable, prepaid expenses, and inventories. "Long-Term Assets" include property, plant, and equipment (PP&E), intangible assets, goodwill, and long-term investments.
- **Liabilities Section:** Liabilities are listed in the order of how close they're to coming due. "Current Liabilities" include accounts payable, accrued expenses, and short-term debt, while "Long-Term Liabilities" include items such as long-term debt, deferred revenue, and deferred income taxes.
- **Shareholders' Equity Section:** The equity section consists of common stock, additional paid-in capital (APIC), treasury stock, and retained earnings.

Could you give further context on what assets, liabilities, and equity each represent?

- **Assets:** Assets are resources with economic value that can be sold for money or bring positive monetary benefits in the future. For example, cash and marketable securities are a store of monetary value that can be invested to earn interest/returns, accounts receivable are payments due from customers, and PP&E is used to generate cash flows in the future – all representing inflows of cash.
- **Liabilities:** Liabilities are unsettled obligations to another party in the future and represent the external sources of capital from third-parties, which help fund the company's assets (e.g., debt capital, payments owed to suppliers/vendors). Unlike assets, liabilities represent future outflows of cash.
- **Equity:** Equity is the capital invested in the business and represents the internal sources of capital that helped fund its assets. The providers of capital could range from being self-funded to outside institutional investors. In addition, the accumulated net profits over time will be shown here as "Retained Earnings."

What are the typical line items you might find on the balance sheet?

Assets Section	
Current Assets <i>(Listed in Order of Liquidity)</i>	
Cash & Cash Equivalents	This line item includes cash itself and highly liquid, cash-like investments, such as commercial paper and short-term government bonds.
Marketable Securities	Marketable securities are short-term debt or equity securities held by the company that can be liquidated to cash relatively quickly.
Accounts Receivable	A/R refers to payments owed to a business by its customers for products and services already delivered to them (i.e., an "IOU" from the customer).
Inventories	Inventories are raw materials, unfinished goods, and finished goods waiting to be sold and the direct costs associated with producing those goods.
Prepaid Expenses	Prepaid expenses are payments made in advance for goods or services expected to be provided on a later date, such as utilities, insurance, and rent.
Non-Current Assets	
Property, Plant & Equipment ("PP&E")	Fixed assets such as land, buildings, vehicles, and machinery used to manufacture or provide the company's services and products.
Intangible Assets	Intangible assets are non-physical, acquired assets such as patents, trademarks, and intellectual property ("IP").
Goodwill	An intangible asset created to capture the excess of the purchase price over the fair market value ("FMV") of an acquired asset.

Liabilities Section	
Current Liabilities <i>(Listed in Order of Liquidity)</i>	
Accounts Payable	A/P represents unpaid bills to suppliers and vendors for services/products already received but were paid for on credit.
Accrued Expenses	Accrued expenses are incurred expenses such as employee compensation or utilities that have not been paid, often due to the invoice not being received.
Short-Term Debt	Debt payments coming due within twelve months, with the current portion of long-term debt also included.
Non-Current Liabilities	
Deferred Revenue	Unearned revenue received in advance for goods or services not yet delivered to the customer (can be either current or non-current).
Deferred Taxes	Tax expense recognized under GAAP but not yet paid because of temporary timing differences between book and tax accounting.
Long-Term Debt	Long-term debt is any debt capital with a maturity exceeding twelve months.
Lease Obligations	Leases are long-term contractual agreements, allowing a company to lease PP&E for a specific time period in exchange for regular payments.

Shareholders' Equity	
Common Stock	Common stock represents a share of ownership in a company and can be issued when raising capital from outside investors in exchange for equity.
Additional Paid-In Capital ("APIC")	APIC represents the amount received in excess over the par value from the sale of preferred or common stock.
Preferred Stock	Preferred stock is a form of equity often considered a hybrid investment, as it has features of both common stock and debt.
Treasury Stock	Refers to shares that had been previously issued but were repurchased by the company in a share buyback and are no longer available to be traded.
Retained Earnings (or Accumulated Deficit)	Represents the cumulative amount of earnings since the company was formed, less any dividends paid out.
Other Comprehensive Income ("OCI")	OCI consists of foreign currency translation adjustments and unrealized gains or losses on available for sale securities.

Apple Inc. | Historical Balance Sheet (Snapshot from [Financial Statement Modeling Course](#))

Apple Balance Sheet (\$ in millions)	2017A 9/30/17	2018A 9/29/18	2019A 12/31/19
Cash & Cash Equivalents	\$20,289	\$25,913	\$48,844
Marketable Securities (Short-Term and Long-Term)	248,606	211,187	157,054
Accounts Receivable, net	17,874	23,186	22,926
Inventories	4,855	3,956	4,106
Other Current Assets	31,735	37,896	35,230
Property, Plant & Equipment, net	33,783	41,304	37,378
Other Non-Current Assets	18,177	22,283	32,978
Total Assets	\$375,319	\$365,725	\$338,516
Accounts Payable	\$44,242	\$55,888	\$46,236
Other Current Liabilities	30,551	33,327	37,720
Deferred Revenue	10,384	5,966	5,522
Commercial Paper	11,977	11,964	5,980
Long-Term Debt (Including Current Portion)	103,703	102,519	102,067
Other Non-Current Liabilities	40,415	48,914	50,503
Total Liabilities	\$241,272	\$258,578	\$248,028
Common Stock & APIC	\$35,867	\$40,201	\$45,174
Retained Earnings	98,330	70,400	45,898
Other Comprehensive Income (OCI)	(150)	(3,454)	(584)
Total Shareholders' Equity	\$134,047	\$107,147	\$90,488

Walk me through the cash flow statement.

There are two methods by which cash flow statements are organized: Direct and Indirect.

The more common approach is the indirect method, whereby the cash flow statement is broken out into three sections:

1. **Cash from Operations:** The cash from operations section starts with net income and adds back non-cash expenses such as depreciation & amortization and stock-based compensation, and then makes adjustments for changes in working capital.
2. **Cash from Investing:** Next, the cash from investing section accounts for capital expenditures (typically the largest outflow), followed by any business acquisitions or divestitures.
3. **Cash from Financing:** In the third section, cash from financing shows the net cash impact of raising capital from issuances of equity or debt, net of cash used for share repurchases, and repayments of debt. The cash outflows from the payout of dividends to shareholders will be reflected here as well.

Together, the sum of the three sections will be the net change in cash for the period. This figure will then be added to the beginning-of-period cash balance to arrive at the ending cash balance.



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The Cash Flow Statement

How are the three financial statements connected?

- **IS ↔ CFS:** The cash flow statement is connected to the income statement through net income, as net income is the starting line on the cash flow statement.
- **CFS ↔ BS:** Next, the cash flow statement is linked to the balance sheet because it tracks the changes in the balance sheet's working capital (current assets and liabilities). The impact from capital expenditures (PP&E), debt or equity issuances, and share buybacks (treasury stock) are also reflected on the balance sheet. In addition, the ending cash balance from the bottom of the cash flow statement will flow to the balance sheet as the cash balance for the current period.
- **IS ↔ BS:** The income statement is connected to the balance sheet through retained earnings. Net income minus dividends issued during the period will be added to the prior period's retained earnings balance to calculate the current period's retained earnings. Interest expense on the income statement is also calculated off the beginning and ending debt balances on the balance sheet, and PP&E on the balance sheet is reduced by depreciation, which is an expense on the income statement.



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3 Financial Statements Connections

If you have a balance sheet and must choose between the income statement or cash flow statement, which would you pick?

Assuming that I would be given both the beginning and end of period balance sheets, I would choose the income statement since I could reconcile the cash flow statement using the balance sheet's year-over-year changes along with the income statement.

Which is more important, the income statement or the cash flow statement?

The income statement and cash flow statement are both necessary, and any in-depth analysis would require using both. However, the cash flow statement is arguably more important because it reconciles net income, the accrual-based bottom line on the income statement, to what is actually occurring to cash.

This means the actual movement of cash during the period is reflected on the cash flow statement. Thus, the cash flow statement brings attention to liquidity-related issues and investments and financing activities that don't show up on the accrual-based income statement.

If you had to pick between either the income statement or cash flow statement to analyze a company, which would you pick?

In most cases, the cash flow statement would be chosen since the cash flow statement reflects a company's true liquidity and is not prone to the same discretionary accounting conventions used in accrual accounting. Whether you're an equity investor or lender, a company's ability to generate sufficient free cash flow to reinvest into its operations and meet its debt obligations comes first. At the end of the day, "cash is king."

Although one factor that could switch the answer is the company's profitability. For an unprofitable company, the income statement can be used to value the company based on a revenue multiple. The cash flow statement becomes less useful for valuation purposes if the company's net income, cash from operations, and free cash flow are all negative.



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Cash Flow
Statement Review

Why is the income statement insufficient to assess the liquidity of a company?

The income statement can be misleading in the portrayal of a company's health from a liquidity and solvency standpoint.

For example, a company can consistently show positive net income yet struggle to collect sales made on credit. The company's inability to retrieve payments from customers would not be reflected on its income statement.

Financial reporting under accrual accounting is also imperfect in the sense that it often relies on management discretion. This "wiggle room" for managerial discretion in reporting decisions increases the risk of earnings management and the misleading depiction of a company's actual operational performance.

Accrual accounting
often relies on
management's
discretion, making the
income statement less
reliable.

The solution to the shortcomings of the income statement is the cash flow statement, which reconciles net income based on the real cash inflows/(outflows) to understand the true cash impact from operations, investing, and financing activities during the period.

What are some discretionary management decisions that could inflate earnings?

- Using excess useful life assumptions for new capital expenditures to reduce the annual depreciation
- Switching from LIFO to FIFO if inventory costs are expected to increase, resulting in higher net income
- Refusing to write-down impaired assets to avoid the impairment loss, which would reduce net income
- Changing policies for costs to be capitalized rather than expensed (e.g., capitalized software costs)
- Repurchasing shares to decrease its share count and artificially increase earnings per share ("EPS")
- Deferral of capex or R&D to the next period to show more profitability and cash flow in the current period
- More aggressive revenue recognition policies in which the obligations of the buyer become less stringent

Tell me about the revenue recognition and matching principle used in accrual accounting.

- **Revenue Recognition Principle:** Revenue is recorded in the same period the good or service was delivered (and therefore "earned"), whether or not cash was collected from the customer.
- **Matching Principle:** The expenses associated with the production/delivery of a good or service must be recorded in the same period as when the revenue was earned.

How does accrual accounting differ from cash-basis accounting?

- **Accrual Accounting:** For accrual accounting, revenue recognition is based on when it's earned and the expenses associated with that revenue are incurred in the same period.

- **Cash-Basis Accounting:** Under cash-basis accounting, revenues and expenses are recognized once cash is received or spent, regardless of whether the product or service was delivered to the customer.

What is the difference between cost of goods sold and operating expenses?

- **Cost of Goods Sold:** COGS represents the direct costs associated with the production of the goods sold or the delivery of services to generate revenue. Examples include direct material and labor costs.
- **Operating Expenses:** Operating expenses such as SG&A and R&D are not directly associated with the production of goods or services offered. Often called indirect costs, examples include rent, payroll, wages, commissions, meal and travel expenses, advertising, and marketing expenses.

When do you capitalize vs. expense items under accrual accounting?

The factor that determines whether an item gets capitalized as an asset or gets expensed in the period incurred is its useful life (i.e., estimated timing of benefits).

- **Capitalized:** Expenditures on fixed and intangible assets expected to benefit the firm for more than one year need to be capitalized and expensed over time. For example, PP&E such as a building can provide benefits for 15+ years and is therefore depreciated over its useful life.
- **Expensed:** In contrast, when the benefits received are short-term, the related expenses should be incurred in the same period. For example, inventory cycles out fairly quickly within a year and employee wages should be expensed when the employee's services were provided.

If depreciation is a non-cash expense, how does it affect net income?

While depreciation is treated as non-cash and an add-back on the cash flow statement, the expense is tax-deductible and reduces the tax burden. The actual cash outflow for the initial purchase of PP&E has already occurred, so the annual depreciation is the non-cash allocation of the initial outlay at purchase.

Do companies prefer straight-line or accelerated depreciation?

For GAAP reporting purposes, most companies prefer straight-line depreciation because lower depreciation will be recorded in the earlier years of the asset's useful life than under accelerated depreciation. As a result, companies using straight-line depreciation will show higher net income and EPS in the initial years.

Eventually, the accelerated approach will show lower depreciation into an asset's life than the straight-line method. However, companies still prefer straight-line depreciation because of the timing, as many companies are focused more on near-term earnings.

If the company is constantly acquiring new assets, the "flip" won't occur until the company significantly scales back capital expenditures.

What is the relationship between depreciation and the salvage value assumption?

Most companies will use a salvage value assumption in which the remaining value of the asset is zero by the end of the useful life. The difference between the cost of the asset and residual value is known as the total depreciable amount. If the salvage value is assumed to be zero, the depreciation expense each year will be higher and the tax benefits from depreciation will be fully maximized.

$$\text{Straight Line Annual Depreciation} = \frac{(\text{Asset Historical Cost} - \text{Salvage Value})}{\text{Useful Life Assumption}}$$

Do companies depreciate land?

While classified as a long-term asset on the balance sheet, land is assumed to have an indefinite useful life under accrual accounting, and therefore depreciation is prohibited.

How would a \$10 increase in depreciation flow through the financial statements?

The depreciation expense will be embedded within either the cost of goods sold or the operating expenses line item on the income statement.

- **IS:** When depreciation increases by \$10, EBIT would decrease by \$10. Assuming a 30% tax rate, net income will decline by \$7.
- **CFS:** At the top of the cash flow statement, net income has decreased by \$7, but the \$10 depreciation will be added back since it's a non-cash expense. The net impact on the ending cash balance will be a positive \$3 increase.
- **BS:** PP&E will decrease by \$10 from the depreciation, while cash will be up by \$3 on the assets side. On the L&E side, the \$7 reduction in net income flows through retained earnings. The balance sheet remains in balance as both sides went down by \$7.

The key takeaway is depreciation, despite being a non-cash expense, reduces taxable income and has a positive impact on the ending cash.

A company acquired a machine for \$5 million and has since generated \$3 million in accumulated depreciation. Today, the PP&E has a fair market value of \$20 million. Under GAAP, what is the value of that PP&E on the balance sheet?

The short answer is \$2 million. Except for certain liquid financial assets that can be written up to reflect their fair market value ("FMV"), companies must carry the value of assets at their net historical cost.

Under IFRS, the revaluation of PP&E to fair value is permitted. Even though permitted, it's not widely used and thus not even well known in the US. Don't voluntarily bring this up in an interview on your own.

What is the difference between growth and maintenance capex?

- **Growth Capex:** The discretionary spending of a business to facilitate new growth plans, acquire more customers, and expand geographically. Throughout periods of economic expansion, growth capex tends to increase across most industries (and the reverse during an economic contraction).
- **Maintenance Capex:** The required expenditures for the business to continue operating in its current state (e.g., repair broken equipment).

Which types of intangible assets are amortized?

Amortization is based on the same accounting concept as depreciation, except it applies to intangible assets rather than fixed tangible assets such as PP&E. Intangible assets include customer lists, copyrights, trademarks, and patents, which all have a finite life and are thus amortized over their useful life.

What is goodwill and how is it created?

Goodwill represents an intangible asset that captures the excess of the purchase price over the fair market value of an acquired business's net assets.

Suppose an acquirer buys a company for a \$500 million purchase price with a fair market value of \$450 million. In this hypothetical scenario, goodwill of \$50 million would be recognized on the acquirer's balance sheet.

Read More → [Goodwill: Differences Between GAAP and Tax Accounting](#)

Goodwill is the excess premium paid over the fair value, and is created to "plug" this gap for the balance sheet to remain in balance.

Can companies amortize goodwill?

Under GAAP, public companies are prohibited from amortizing goodwill as it's assumed to have an indefinite life, similar to land. Instead, goodwill must be tested annually for impairment.

However, privately held companies may elect to amortize goodwill and under some circumstances, goodwill can be amortized over 15 years for tax reporting purposes.

What is the “going concern” assumption used in accrual accounting?

In accrual accounting, companies are assumed to continue operating into the foreseeable future and remain in existence indefinitely. The assumption has broad valuation implications, given the expectation of continued cash flow generation from the assets belonging to a company, as opposed to being liquidated.

Explain the reasoning behind the principle of conservatism in accrual accounting.

The conservatism principle requires thorough verification and use of caution by accountants when preparing financial statements, which leads to a downward measurement bias in their estimates.

Central to accounting conservatism is the belief that it's better to understate revenue or the value of assets than to overstate it (and the reverse for expenses and liabilities). As a result, the risk of a company's revenue or asset values being overstated and expenses or liabilities being understated is minimized.

For any revenue or expense to be recognized, there must be evidence of occurrence with a measurable monetary amount.

Why are most assets recorded at their historical cost under accrual accounting?

The historical cost principle states that an asset's value on the balance sheet must reflect the initial purchase price, not the current market value. This guideline represents the most consistent measurement method since there's no need for constant revaluations and markups, thereby reducing market volatility.

What role did fair-value accounting have in the subprime mortgage crisis?

In the worst-case scenario, sudden drops in asset values could cause a domino effect in the market. An example was the subprime mortgage crisis, in which the meltdown's catalyst is considered to be FAS-157. This mark-to-market accounting rule mandated financial institutions to update their pricing of illiquid securities. Soon after, write-downs in financial derivatives, most notably credit default swaps ("CDS") and mortgage-backed securities ("MBS"), ensued from commercial banks, and it was all downhill from there.

Why are the values of a company's intangible assets not reflected on its balance sheet?

The objectivity principle of accrual accounting states that only verifiable, unbiased data can be used in financial filings, as opposed to subjective measures. For this reason, internally developed intangible assets such as branding, trademarks, and intellectual property will have no value recorded on the balance sheet because they cannot be accurately quantified and recorded.

Companies are not permitted to assign values to these intangible assets unless the value is readily observable in the market via acquisition. Since there's a confirmable purchase price, a portion of the excess amount paid can be allocated towards the rights of owning the intangible assets and recorded on the closing balance sheet.

If the share price of a company increases by 10%, what is the balance sheet impact?

There would no change on the balance sheet as shareholders' equity reflects the book value of equity. Equity value, also known as the "market capitalization," represents the value of a company's equity based on supply and demand in the open market. In contrast, the book value of equity is the initial historical amount shown on the balance sheet for accounting purposes. This represents the company's residual value belonging to equity shareholders once all of its assets are liquidated and liabilities are paid off.



**Watch Video →
Book v Market Value**

$$\text{Book Value of Equity} = \text{Total Assets} - \text{Total Liabilities}$$

The equity value recorded on the books will be significantly understated from the market value in most cases.

For example, the book value of Apple's common stock is only ~\$51 billion as of its latest 10-K filing for FY 2020, whereas its market value of equity is over \$2 trillion as of this guide's publishing date.

Do accounts receivable get captured on the income statement?

There is no accounts receivable line item on the income statement, but it gets captured, if only partially, indirectly in revenue. Under accrual accounting, revenue is recognized during the period it was earned, whether or not cash was received.

The two other financial statements would be more useful to understand what is happening to the accounts receivable balance since the cash flow statement will reconcile revenue to cash revenue, while the absolute balance of accounts receivable can be observed on the balance sheet.

Why are increases in accounts receivable a cash reduction on the cash flow statement?

Since the cash flow statement begins with net income and net income captures all of a company's revenue (not just cash revenue), an increase in accounts receivable means that more customers paid on credit during the period.

Thus, a downward adjustment must be made to net income to arrive at the ending cash balance. Although the revenue has been earned under accrual accounting standards, the customers have yet to make the due cash payments and this amount will be sitting as receivables on the balance sheet.

What is deferred revenue?

Deferred revenue (or "unearned" revenue) is a liability that represents cash payments collected from customers for products or services not yet provided. Some examples are gift cards, service agreements, or implied rights to future software upgrades associated with a product sold. In all the examples listed, the cash payment was received upfront and the benefit to the customer will be delivered on a later date.



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Deferred Revenue

For instance, a company that sells a smartphone for \$500 might allocate \$480 of the sale to the phone and the remaining \$20 to the value of the customer's right to future software upgrades. Here, the company would collect \$500 in cash, but only \$480 would be recognized as revenue. The remaining \$20 will stay recognized as deferred revenue until the software upgrades are provided.

Why is deferred revenue classified as a liability while accounts receivable is an asset?

- **Deferred Revenue:** For deferred revenue, the company received payments upfront and has unfulfilled obligations to the customers that paid in advance, hence its classification as a liability.
- **Accounts Receivable:** A/R is an asset because the company has already delivered the goods/services and all that remains is the collection of payments from the customers that paid on credit.

Why are increases in accounts payable shown as an increase in cash flow?

An increase in accounts payable would mean the company has been delaying payments to its suppliers or vendors, and the cash is currently still in the company's possession. The due payments will eventually be made, but the cash belongs to the company for the time being and is not restricted from being used. Thus, an increase in accounts payable is reflected as an inflow of cash on the cash flow statement.

Which section of the cash flow statement captures interest expense?

The cash flow statement doesn't directly capture interest expense. However, interest expense is recognized on the income statement and then gets indirectly captured in the cash from operations section since net income is the starting line item on the cash flow statement.

What happens to the three financial statements if a company initiates a dividend?

- **IS:** When a company initiates a dividend, there'll be no changes to the income statement. However, a line below net income will state the dividend per share ("DPS") to show the amount paid.
- **CFS:** On the cash flow statement, the cash from financing section will decrease by the dividend payout amount and lower the ending cash balance at the bottom.
- **BS:** The cash balance will decline by the dividend amount on the balance sheet, and the offsetting entry will be a decrease in retained earnings since dividends come directly out of retained earnings.

Do inventories get captured on the income statement?

There is no inventory line item on the income statement, but it gets indirectly captured, if only partially, in cost of goods sold (or operating expenses). For a specific period, regardless of whether the associated inventory was purchased during the same period, COGS may reflect a portion of the inventory used up.

The two other financial statements would be more useful for assessing inventory as the cash flow statement shows the year-over-year changes in inventory, while the balance sheet shows the beginning and end-of-period inventory balances.

How should an increase in inventory get handled on the cash flow statement?

An increase in inventory reflects a use of cash and should thus be reflected as an outflow on the cash from operations section of the cash flow statement. The inventory balance increasing from the prior period implies the amount of inventory purchased exceeded the amount expensed on the income statement.

What is the difference between LIFO and FIFO, and what are the implications on net income?

FIFO and LIFO are two inventory accounting methods to estimate the value of inventory sold in a period.

- **First In, First Out ("FIFO"):** Under FIFO accounting, the goods that were purchased earlier would be the first ones to be recognized and expensed on the income statement.
- **Last In, First Out ("LIFO"):** Alternatively, LIFO assumes that the most recently purchased inventories are recorded as the first ones to be sold first.

The impact on net income would depend on how inventory costs have changed over time:

	First In, First Out (FIFO)	Last In, First Out (LIFO)
Rising Inventory Costs (↑)	▪ If inventory costs have been rising, lower COGS would be recorded under FIFO. ▪ Since the less expensive inventory was recognized, net income will be higher in the current period.	▪ If inventory costs have been rising, then COGS for the period will be higher under LIFO because the recent, pricier purchases are assumed to be sold first. ▪ Thus, the result would be lower net income for the period.
Decreasing Inventory Costs (↓)	▪ If inventory costs have been dropping, COGS would be higher under FIFO, since older inventory costs are more expensive. ▪ The ending result would be lower net income for the period.	▪ If inventory costs have been dropping, then COGS would be lower under LIFO. ▪ Thereby, net income for the period will be higher since the cheaper inventory costs were recognized.

What is the average cost method of inventory accounting?

Besides FIFO and LIFO, the average cost method is the third most widely used inventory accounting method. Under this method, the assigned inventory costs are based on a weighted average, in which the total costs of production in a period are summed up and divided by the total number of items produced.

Each product cost is treated equally and inventory costs are spread out evenly, disregarding the date of purchase or production. Thus, this method is viewed as a simplistic compromise between the two other methods, but would be improper to use if the products sold are each unique with significant variance in the cost to manufacture and the sale price (i.e., more applicable for high-volume, identical batches of inventory).

How do you calculate retained earnings for the current period?

Retained earnings represent the total cumulative amount of net income held onto by a company since inception after accounting for any dividends paid out to its common and preferred shareholders.

$$\text{Current Period Retained Earnings} = \text{Prior Retained Earnings} + \text{Net Income} - \text{Dividends}$$

What does the retention ratio represent and how is it related to the dividend payout ratio?

The retention ratio represents the proportion of net income retained by the company, net of any dividends paid out to shareholders. The inverse of the retention ratio is the dividend payout ratio, which measures the proportion of net income paid out as dividends to investors.

$$\text{Retention Ratio} = \frac{(\text{Net Income} - \text{Dividends})}{\text{Net Income}} \quad \text{Dividend Payout Ratio} = \frac{\text{Dividends Paid}}{\text{Net Income}}$$

What are the two ways to calculate earnings per share (EPS)?

1. **Basic EPS:** Determines a company's earnings on a per-share basis, but allocable to only the basic shares outstanding (otherwise known as "common shares").

$$\text{Basic EPS} = \frac{(\text{Net Income} - \text{Dividends on Preferred Stock})}{\text{Basic Weighted Average Shares Outstanding}}$$

2. **Diluted EPS:** Compares a company's earnings relative to its shares outstanding on a per-share basis but considers the impact of potentially dilutive securities such as options, warrants, and convertible securities. If an option is "in-the-money," the option holder can become a common shareholder at their choosing. Thus, diluted EPS is a more accurate depiction of ownership value per share.

$$\text{Diluted EPS} = \frac{(\text{Net Income} - \text{Dividends on Preferred Stock})}{\text{Diluted Weighted Average Shares Outstanding}}$$

Read More → [Forecasting Shares Outstanding and Earnings Per Share \(EPS\)](#)

Where can you find the financial reports of public companies?

In the US, public companies are required to report periodic filings with the SEC, including an annual report (10-K) and three quarterly (10-Q) reports each year. These reports are available for free through [SEC EDGAR](#).

In other countries, reporting requirements will vary, but most countries will require at least an annual report, while some will require an interim filing (i.e., a report in the middle of the company's fiscal year). Only a few countries make company filings easily accessible through a central database, forcing analysts to rely on expensive financial data providers or dig through company websites manually to collect data.

The closest database to EDGAR in breadth and ease of use is Canada's [SEDAR database](#). In the United Kingdom, the closest EDGAR equivalent is [Companies House](#), where private companies must also report their financials.

Read More → [Bloomberg vs. Capital IQ vs. FactSet vs. Thomson Reuters Eikon](#)

What is a proxy statement?

The proxy statement, formally known as "Form 14A," is required to be filed before a shareholder meeting to solicit shareholder votes. The document must disclose all relevant details regarding the matter for shareholders to make an informed decision.

In addition, the board of directors' compensation and other notable announcements such as changes to the company's articles of incorporation are included.

What is an 8-K and when is it required to be filed?

An 8-K is a required filing with the SEC when a company undergoes a materially significant event and must disclose the details. Often called the "current report," 8-Ks are usually filed within four days of the event. The information contained within the report should be of high importance and pertinent to shareholders.

Events that would trigger this filing would include previously unannounced plans for a new acquisition, disposal of assets, bankruptcy, a tender offer, the resignation of a senior-level manager or member of the board of directors, or disclosure that the company is under SEC investigation for alleged wrongdoing.

Why has understanding the differences between US GAAP and IFRS financial reporting become increasingly important?

For public companies in the US, the reporting rules and guidelines are set by the Financial Accounting Standards Board (FASB) and referred to as US Generally Accepted Accounting Principles (US GAAP). The International Accounting Standards Board (IASB) oversees the International Financial Reporting Standards (IFRS), which is followed by over 144 countries.

Understanding the differences between US GAAP and IFRS has become more important because:

- **Continuation of Globalization:** Globalization is the gradual convergence of economies in different countries. The widespread adoption of IFRS has placed pressure on the US to adopt IFRS to have a single set of accounting standards and rules used worldwide, but it seems unlikely in the near-term.
- **Geographic Diversification of Investments:** In recent years, investment firms have been broadening their investments' geographic scope to consider more opportunities overseas. Nowadays, institutional investors are more open to making investments in emerging markets due to the prevalence of opportunities and as a strategy to re-risk their overall portfolio.
- **Cross-Border M&A Activity:** Cross-border mergers and acquisitions ("M&A") have emerged as a strategy for multinational companies to enter new markets, extend their reach to new potential customers, and diversify their revenue sources.

Financial Statement Analysis

What are some of the most common margins used to measure profitability?

Profitability Margins		
Gross Margin	The percentage of revenue remaining after subtracting just COGS, the direct costs associated with the company's revenue generation (e.g., direct materials, direct labor).	$\text{Gross Margin} = \frac{\text{Gross Profit}}{\text{Revenue}}$
Operating Margin	The percentage of profitability after subtracting operating expenses from gross profit. This measure is useful for comparisons due to being independent of capital structure and taxes.	$\text{Operating Margin} = \frac{\text{EBIT}}{\text{Revenue}}$
Net Profit Margin	The percentage of accrual profitability remaining after all expenses have been subtracted. Unlike operating margin, this measure is impacted by capital structure and taxes.	$\text{Net Profit Margin} = \frac{\text{Net Income}}{\text{Revenue}}$
EBITDA Margin	The most widely used profit margin for benchmarking due to being independent of capital structure and taxes, in addition to adjusted for non-cash expenses (D&A) and non-recurring items.	$\text{EBITDA Margin} = \frac{\text{EBITDA}}{\text{Revenue}}$

What do the phrases “above the line” and “below the line” mean?

The expression “the line” is in reference to operating income, which represents the point that divides normal, ongoing business operations from non-operational line items.

- **“Above the Line”:** If a profitability metric is “above the line,” it reflects a company’s operational performance before non-operational items such as interest and taxes. Financing-related activities are an example of such non-operational items, as decisions on how to fund a company are discretionary (debt vs. equity). For example, a metric such as earnings before interest, taxes, depreciation and amortization (“EBITDA”) is considered “above the line.” Hence, its widespread usage for comparative purposes since operational performance is portrayed while being independent of capital structure and taxes.
- **“Below the Line”:** In contrast, profitability metrics “below the line” have adjusted operating income for non-operating income and expenses, which are items classified as discretionary and unrelated to the core operations of a business. An example would be net income, since interest expense, non-operating income/(expenses), and taxes have all been accounted for in its ending value.

Is EBITDA a good proxy for operating cash flow?

While EBITDA does add back D&A, typically the largest non-cash expense, it doesn't capture the full cash impact of capital expenditures (“capex”) or working capital changes during the period.

EBITDA also doesn't adjust for stock-based compensation, although an increasingly used “adjusted EBITDA” metric does add-back SBC. These non-cash and any non-recurring adjustments must be properly accounted for to assess a company's past operational performance and to accurately forecast its future cash flows.

Read More → [EBITDA vs. Cash Flows from Operations vs. Free Cash Flow](#)

Despite the criticism regarding its drawbacks, EBITDA remains the most widely used proxy for operating cash flow in practice.

What are some examples of non-recurring items?

Non-recurring items include legal settlements (gain or loss), restructuring expenses, inventory write-downs, or asset impairments. Often called "scrubbing" the financials, the act of adjusting for these non-recurring items is meant to normalize the cash flows and depict a company's true operating performance.

When adjusting for non-recurring expenses, are litigation expenses always added back?

Not necessarily, as whether an expense is non-recurring depends on the industry. In many cases, it's a discretionary decision on whether an expense is a part of the normal operations of a company. For example, expenses related to litigation might not be added back for a research and development (R&D) oriented pharmaceutical company, given the prevalence of lawsuits in the industry.

What is the difference between organic and inorganic revenue growth?

- **Organic Growth:** A company experiencing organic growth is expanding to new markets, enhancing its sales & marketing strategies, improving its product/service mix, or introducing new products. The focus is on continuously making operational improvements and bringing in revenue (e.g., set prices more appropriately post-market research, target right end markets).
- **Inorganic Growth:** Once the opportunities for organic growth have been maximized, a company may turn to inorganic growth, which refers to growth driven by M&A. Inorganic growth is often considered faster and more convenient than organic growth. Post-acquisition, a company can benefit from synergies, such as having new customers to sell to, bundling complementary products, and diversification in revenue.

Simply put, organic growth is the optimization of a business from the internal efforts of management and their employees.

How does the relationship between depreciation and capex shift as companies mature?

The more a company has spent on capex in recent years, the more depreciation the company incurs in the near-term future. Therefore, when looking at high-growth companies spending heavily on growth capex, their ratio between capex and annual depreciation will far exceed 1.

For mature businesses experiencing stagnating or declining growth, this ratio converges near 1, as the only capex is related to routine maintenance capex (e.g., replace equipment, refurbish store layouts).

As a company begins to mature and its growth stagnates, a greater proportion of its total capex will shift towards maintenance.

What is working capital?

The working capital metric measures a company's liquidity and ability to pay off its current obligations using its current assets. In general, the more current assets a company has relative to its current liabilities, the lower its liquidity risk. Current liabilities represent payments that a company needs to make within the year (e.g., accounts payable, accrued expenses), whereas current assets are resources that can be turned into cash within the year (e.g., accounts receivable, inventory).

$$\text{Working Capital} = \text{Current Assets} - \text{Current Liabilities}$$

Read More → [Working Capital: Formulas, Misconceptions and Real Examples](#)

Why are cash and debt excluded in the calculation of net working capital (NWC)?

In practice, cash and other short-term investments (e.g., treasury bills, marketable securities, commercial paper) and any interest-bearing debt (e.g., loans, revolver, bonds) are excluded when calculating working capital because they're non-operational and don't directly generate revenue.

$$\text{Net Working Capital (NWC)} = \text{Operating Current Assets} - \text{Operating Current Liabilities}$$

Cash & cash equivalents are closer to investing activities since the company can earn a slight return (~0.25% to 1.5%) through interest income, whereas debt is classified as financing. Neither is operations-related, and both are thereby excluded in the calculation of NWC.

Read More → [Working Capital in Valuation](#)

Is negative working capital a bad signal about a company's health?

Further context would be required, as negative working capital can be positive or negative. For instance, negative working capital can result from being efficient at collecting revenue, quick inventory turnover, and delaying payments to suppliers while efficiently investing excess cash into high-yield investments.

However, the opposite could be true, and negative working capital could signify impending liquidity issues. Imagine a company that has mismanaged its cash and faces a high accounts payable balance coming due soon, with a low inventory balance that desperately needs replenishing and low levels of AR. This company would need to find external financing as early as possible to stay afloat.

What does change in net working capital tell you about a company's cash flows?

The change in net working capital is important because it gives you a sense of how much a company's cash flows will deviate from its accrual-based net income.

$$\text{Change in Net Working Capital} = \text{NWC}_{\text{Prior Period}} - \text{NWC}_{\text{Current Period}}$$

If a company's NWC has increased year-over-year, its operating assets have grown and/or its operating liabilities have shrunk from the prior year. Since an increase in an operating asset is a cash outflow, it should be intuitive why an increase in NWC means less cash flow for a company (and vice versa).

What ratios would you look at to assess working capital management efficiency?

Efficiency Ratios		
Days Inventory Held ("DIH")	- DIH measures the average number of days it takes for a company to sell off its inventory. - Companies strive to minimize their DIH and sell their inventory as soon as possible.	$\text{DIH} = \left(\frac{\text{Inventory}}{\text{COGS}} \right) \times 365 \text{ Days}$
Days Sales Outstanding ("DSO")	- DSO represents the average number of days it takes for a company to collect payments made on credit. - Lower DSOs mean less time is needed to retrieve cash from sales made on credit.	$\text{DSO} = \left(\frac{\text{AR}}{\text{Revenue}} \right) \times 365 \text{ Days}$
Days Payable Outstanding ("DPO")	- DPO refers to the average number of days it takes for a company to pay back its suppliers. - Higher DPOs indicate the company has more bargaining power over its suppliers.	$\text{DPO} = \left(\frac{\text{AP}}{\text{COGS}} \right) \times 365 \text{ Days}$

What is the cash conversion cycle?

The cash conversion cycle ("CCC") measures the number of days it takes a company to convert its inventory into cash from sales. Therefore, a lower cash conversion cycle is preferred as it implies the company generates and collects cash in a shorter duration. As a general rule, companies with lower CCCs operate efficiently, hold more negotiating power over suppliers, and have quicker sales collection cycles.

$$\text{Cash Conversion Cycle} = \text{DIO} + \text{DSO} - \text{DPO}$$

How would you forecast working capital line items on the balance sheet?

Working Capital Forecast		
Accounts Receivable (A/R)	- A/R typically grows along with revenue and should remain in-line with historical periods. - DSO often gradually decreases over time as companies become more efficient in payment collection.	- DSO should be calculated for historical periods ($DSO = A/R \div \text{Revenue} \times 365$) - Assumptions for DSO in future periods will be based on historical trends or an average. - Forecasted A/R can be calculated by $(DSO \text{ assumption} \div 365) \times \text{forecasted revenue}$.
Inventories	- Inventory will grow in-line with COGS in most cases. - DIH can decrease over time as companies become more efficient at selling their inventory.	- DIH would be calculated for historical periods ($DIH = \text{Inventory} \div \text{COGS} \times 365$) - Historical trends or an average of past periods should be used for the DIH assumptions. - Forecasted inventory balance will be equal to the $(DIH \text{ assumption} \div 365) \times \text{forecasted COGS}$.
Prepaid Expenses	Prepaid expenses will typically be SG&A related, but can grow with revenue if it's unclear.	- Historical prepaid expenses would be calculated as a % of SG&A (or revenue). - Forecasted prepaid expenses will be equal to the % assumption $\times$ forecasted revenue.
Other Current Assets	- Other current assets normally grow along with revenue, assuming they're tied to operations. - Otherwise, can be straight-lined	- Other current assets for past periods would be calculated by dividing past amounts by revenue. - Forecasted OCA would be equal to the % assumption $\times$ forecasted revenue.
Accounts Payable (A/P)	- A/P will typically grow with COGS, especially if the company sells goods (i.e., inventory payment delays). - DPO can be assumed to gradually increase if the company might have more buyer power in the future.	- DPO for past periods will be calculated ($DPO = A/P \div \text{COGS} \times 365$). - Historical trends would be followed or an average can be taken for the assumption. - Forecasted A/P will be equal to the $(DPO \text{ assumption} \div 365) \times \text{COGS}$.
Accrued Expenses	Accrued expenses usually relate to operating expenses, thus it will grow along with SG&A (or revenue if it's unclear).	- Accrued expenses as a percentage of SG&A will first be calculated for historical periods. - Forecasted accrued expenses will be the % SG&A assumption $\times$ current period SG&A.
Deferred Revenue	Deferred revenue is typically forecasted to grow with the revenue, as it's likely a part of the company's business model.	- Deferred revenue as a % of revenue will be calculated for historical periods. - Forecasted D/R will be equal to the % of revenue assumption $\times$ forecasted revenue.
Other Current Liabilities	- Current liabilities are usually forecasted to grow with revenue - If the driver is unclear, it can be straight-lined.	- Current liabilities as a % of revenue will be calculated for previous periods. - Forecasted OCL will be equal to the % of revenue assumption $\times$ current period revenue.

How would you forecast capex and D&A when creating a financial model?

In the simplest approach, D&A can be projected as either a percentage of revenue or capital expenditures, while capex is forecasted as a percentage of revenue. Re-investments such as capex directly correlate with revenue growth, thus historical trends, management guidance, and industry norms should be closely followed.

Alternatively, a depreciation waterfall schedule can be put together, which would require more data from the company to track the PP&E currently in-use and the remaining useful life of each. In addition, management plans for future capex spending and the approximate useful life assumptions for each purchase will be necessary. As a result, depreciation from old and new capex will be separately shown.

For projecting amortization, useful life assumptions would also be required, which can often be found in a separate footnote in a company's financial reports.

How would you forecast PP&E and intangible assets?

When forecasting PP&E, the end of period balance will be calculated using the roll-forward schedule shown below. Note, capex will input as a negative, meaning the PP&E balance should increase. Other factors that could affect the end-of-period PP&E balance are asset sales and write-downs.

PP&E Roll-Forward

$$\text{EOP PP\&E} = \text{BOP PP\&E} + \text{Capex} - \text{Depreciation}$$

To forecast intangible assets, management guidance becomes necessary as unlike capex, there's usually no clear historical pattern that can be followed as these purchases tend to be inconsistent. In most cases, it's best to rely on management if available, but in the absence of guidance, it's recommended to assume no purchases.

Intangible Assets Roll-Forward

$$\text{EOP Intangibles} = \text{BOP Intangibles} + \text{Intangibles Purchases} - \text{Amortization}$$

What is the difference between the current ratio and the quick ratio?

The current ratio and quick ratio are used to assess a company's near-term liquidity position. The two ratios are both used to determine if a company can meet its short-term obligations using just its short-term assets at the present moment.

- **Current Ratio:** A current ratio greater than 1 implies that the company is financially healthy in terms of liquidity and can meet its short-term obligations.

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

- **Quick Ratio:** Otherwise known as the acid-test ratio, the quick ratio measures short-term liquidity, but uses stricter policies on what classifies as a liquid asset. Therefore, it includes only highly liquid assets that could be converted to cash in less than 90 days with a high degree of certainty.

$$\text{Quick Ratio} = \frac{(\text{Cash \& Cash Equivalents} + \text{AR} + \text{Short Term Investments})}{\text{Current Liabilities}}$$

Give some examples of when the current ratio might be misleading?

- The cash balance used includes the minimum cash amount required for working capital needs – meaning operations could not continue if cash were to dip below this level.
- Similarly, the cash balance may contain restricted cash, which is not freely available for use by the business and is instead held for a specific purpose.
- Short-term investments that cannot be liquidated in the markets easily could have been included (i.e., low liquidity, cannot sell without a substantial discount).
- Accounts receivable could include “bad A/R”, but management refuses to recognize it as such.

Is it bad if a company has negative retained earnings?

Not necessarily. Retained earnings can turn negative if the company has generated more accounting losses than profits. For example, this is often the case for startups and early-stage companies investing heavily to support future growth (e.g., high capex, sales & marketing expenses, R&D spend).

Another component of retained earnings is the payout of dividends and share repurchases, contributing to lower or even negative retained earnings. In these scenarios, the negative retained earnings mean the company has returned more capital to shareholders than taken in.

How can a profitable firm go bankrupt?

To be profitable, a company must generate revenues that exceed expenses. However, if the company is ineffective at collecting cash from customers and allows its receivables to balloon, or if it cannot get favorable terms from suppliers and must pay cash for all inventories and supplies, the company can suffer from liquidity problems due to the timing mismatch of cash inflows and outflows.

Profitable companies with these types of working capital issues can usually secure financing, but if financing suddenly becomes unavailable (e.g., 2008 credit crisis), the company could be forced to declare bankruptcy.

Alternatively, a profitable company that took on far too much debt in its capital structure and could not service the interest payments may also default on its debt obligations.

What does return on assets (ROA) and return on equity (ROE) each measure?

Return on assets ("ROA") and return on equity ("ROE") are measures of profitability that show how effective a company's management team is at utilizing the resources it has on-hand (assets or equity).

- **Return on Assets:** ROA measures asset utilization and how efficiently a company's assets are used to generate earnings. A high ROA relative to a peer group indicates assets are being used near full capacity, whereas a low ROA means management may not be deriving the full potential benefit from its assets.

$$\text{Return on Assets (ROA)} = \frac{\text{Net Income}}{(\text{Average of Beginning and Ending Total Assets})}$$

- **Return on Equity:** The ROE ratio gives insight into how efficiently a management team has been using the capital shareholders have contributed. A higher ROE means management is efficient at using the money raised from equity financing (and vice versa).

$$\text{Return on Equity (ROE)} = \frac{\text{Net Income}}{(\text{Average of Beginning and Ending Book Value of Equity})}$$

What is the relationship between return on assets (ROA) and return on equity (ROE)?

The relationship between ROA and ROE is tied to the use of leverage. In the absence of debt in the capital structure, the two metrics would be equal. But if the company were to add debt to its capital structure, its ROE would rise above its ROA due to increased cash, as total assets would rise while equity decreases.

If a company has a ROA of 10% and a 50/50 debt-to-equity ratio, what is its ROE?

Imagine a company with \$100 in total assets. A 10% return on assets (ROA) would imply \$10 in net income. Since the debt-to-equity mix is 50/50, the return on equity (ROE) is \$10/\$50 = 20%.

When using metrics such as ROA and ROE, why do we use averages for the denominator?

The numerator, usually net income, comes from the income statement. The denominator, either assets or equity, comes from the balance sheet. The income statement covers a specific period, whereas the balance sheet is a snapshot at one particular point in time. Thus, the average between the beginning and ending balance of the denominator is used to adjust for this mismatch in timing.

What are some shortcomings of the ROA and ROE metrics for comparison purposes?

A company's ROA and ROE ratios are benchmarked against competitors in the same industry to assess management efficiency and track historical trends. However, the ROA and ROE ratios are most useful when compared to a peer group of companies with similar growth rates, margin profiles, and risks. This approach would be best suited for established companies operating in mature, low-growth industries with many comparable companies to accurately track the management team's profitability and efficiency.

What is the return on invested capital (ROIC) metric used to measure?

The return on invested capital ("ROIC") metric is used to assess how efficient a management team is at capital allocation. A company that generates an ROIC over its cost of capital (WACC) suggests the management team has been allocating capital efficiently (i.e., investing in profitable projects or investments) and if sustained over the long-run, this indicates a competitive advantage. ROIC represents one of the most fundamental assessments of a company: "How much in returns is the company earning for each dollar invested?"

$$\text{Return on Invested Capital (ROIC)} = \frac{\text{NOPAT}}{\text{Invested Capital}}$$

Read More → [Calculating Return on Invested Capital](#)

What does the asset turnover ratio measure?

The asset turnover ratio is a metric used to understand how efficiently a company uses its assets to generate sales. The asset turnover ratio answers the question, "How many dollars in revenue does the company generate per dollar of assets?" The higher the ratio, the better, as this suggests the company is generating more revenue per dollar of an asset owned. But it has shortfalls in being distorted by capital expenditures and asset sales.

$$\text{Asset Turnover Ratio} = \frac{\text{Revenue}}{(\text{Average of Beginning and Ending Total Assets})}$$

What does inventory turnover measure and how does it differ from days inventory held (DIH)?

The inventory turnover ratio is how often a company has sold and replaced its inventory balance throughout a specified period (i.e., the number of times inventory was "turned over").

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{(\text{Average of Beginning and Ending Inventory})}$$

In contrast, DIH is the average number of days it takes for a company to turn its inventory into revenue.

What does accounts receivables turnover measure?

Accounts receivable turnover is a metric used to measure the number of times per year that a company can collect its average accounts receivable from customers. The higher the turnover ratio, the better as it indicates the company is efficient at collecting its due payments from customers that paid on credit.

$$\text{Receivables Turnover} = \frac{\text{Revenue}}{(\text{Average of Beginning and Ending Accounts Receivables})}$$

What does accounts payables turnover measure and is a higher or lower number preferable?

Accounts payable turnover measures how quickly a company pays its vendors. Generally, longer credit terms provide a company with more flexibility as it means the company has more cash-on-hand. A higher A/P turnover means the company pays off its A/P balance quickly, meaning the cash outflows occur faster.

$$\text{Accounts Payable Turnover} = \frac{\text{Cost of Goods Sold}}{(\text{Average Beginning and Ending Accounts Payable})}$$

What are some ratios you would look at to perform credit analysis?

Credit Ratio Type	Purpose	Examples
Liquidity Ratios	Assesses a company's ability to meet its current obligations using its current assets.	- Current Ratio - Quick Ratio - Cash Ratio
Leverage/Solvency Ratios	Compares a company's use of debt to assets, equity, earnings, and total capitalization to evaluate if its debt obligations can be met.	- Debt-to-EBITDA - Debt-to-Assets - Debt-to-Equity
Coverage Ratios	Measures a company's ability to service interest payments and other debt-related obligations using a cash flow metric.	- Times Interest Earned - EBITDA Interest Coverage Ratio - Debt Service Coverage Ratio - Fixed Charge Coverage Ratio
Profitability Ratios	Conveys a company's ability to consistently generate profits, enabling it to meet its debt obligations.	- Gross, Operating, Net Profit, and EBITDA Margins % - ROE, ROA, and ROIC

What are the two types of credit ratios used to assess a company's default risk?

Credit Ratios		
Leverage Ratios	Leverage ratios compare the amount of debt held by a company to a specific cash flow metric, most often EBITDA.	- Total Debt/EBITDA - Senior Debt/EBITDA - Net Debt/EBITDA - Total Debt/Equity - Total Debt/Total Capital
Interest Coverage Ratios	Interest coverage ratios measure a company's ability to cover its interest payments using its cash flows. The higher the interest coverage ratio, the better (ideally, >2.0x).	- EBIT/Interest Expense - EBITDA/Interest Expense - EBITDA/Cash Interest Expense (EBITDA – Capex)/Interest Expense

How do you calculate the debt service coverage ratio (DSCR) and what does it measure?

The debt service coverage ratio (DSCR) is a measure of creditworthiness that tests a company's ability to pay its current debt obligations using its current cash flows. As a general rule, a DSCR greater than 1.0 shows the company is generating sufficient cash flows to pay down its debt. But a DSCR below 1.0 could be a cause of concern, as it suggests the company might have insufficient cash flows to handle the debt it currently holds.

There are various methods to calculate the DSCR, but one commonly used example is shown below:

$$\text{DSCR} = \frac{(\text{EBITDA} - \text{Capex})}{(\text{Mandatory Principal Repayment} + \text{Interest Expense})}$$

How do you calculate the fixed charge coverage ratio (FCCR) and what does it mean?

The fixed charge coverage ratio (FCCR) is used to assess if a company's earnings can cover its fixed charges, which can include rent, utilities, and interest expense. The higher the ratio, the better the creditworthiness. Fixed charges can include expenses such as rent or lease payments, and utility bills.

$$\text{Fixed Charge Coverage Ratio (FCCR)} = \frac{(\text{EBIT} + \text{Lease Charges})}{(\text{Lease Charges} + \text{Interest Expense})}$$

Advanced Accounting

How would raising capital through share issuances affect earnings per share (EPS)?

The impact on EPS is that the share count increases, which decreases EPS. But there can be an impact on net income, assuming the share issuances generate cash because there would be higher interest income, which increases net income and EPS. However, most companies' returns on excess cash are low, so this doesn't offset the negative dilutive impact on EPS from the increased share count.

Alternatively, share issuances might affect EPS in an acquisition where stock is the form of consideration. The amount of net income the acquired company generates will be added to the acquirer's existing net income, which could have a net positive (accretive) or negative (dilutive) impact on EPS.

How would a share repurchase impact earnings per share (EPS)?

The impact on EPS following a share repurchase is a reduced share count, which increases EPS. However, there would be an impact on net income, assuming the share repurchase was funded using excess cash. The interest income that would have otherwise been generated on that cash is no longer available, causing net income and EPS to decrease.

But the impact would be minor since the returns on excess cash are low, and would not offset the positive impact the repurchase had on EPS from the reduced share count.

What is the difference between the effective and marginal tax rates?

- **Effective Tax Rate:** The effective tax rate represents the percentage of taxable income corporations must pay in taxes. For historical periods, the effective tax rate can be backed out by dividing the taxes paid by the pre-tax income (or earnings before tax).

$$\text{Effective Tax Rate \%} = \frac{\text{Taxes Paid}}{\text{Earnings Before Tax}}$$

- **Marginal Tax Rate:** The marginal tax rate is the taxation percentage on the last dollar of a company's taxable income. The tax expense depends on the statutory tax rate of the governing jurisdiction and the company's taxable income, as the tax rate adjusts according to the tax bracket in which it falls.

Why is the effective and marginal tax rate often different?

Effective and marginal tax rates differ because the effective tax rate calculation uses pre-tax income from the accrual-based income statement. Since there's a difference between the taxable income on the income statement and taxable income shown on the tax filing, the tax rates will nearly always be different. Thus, the "Tax Provision" line item on the income statement rarely matches the actual cash taxes paid to the IRS.

Read More → [Effective vs. Marginal Tax Rates](#)

Could you give specific examples of why the effective and marginal tax rates might differ?

Under GAAP, many companies follow different accounting standards and rules for tax and financial reporting.

- Most companies use straight-line depreciation (i.e., equal allocation of the expenditure over the useful life) for reporting purposes, but the IRS requires accelerated depreciation for tax purposes – meaning, book depreciation is lower than tax depreciation for earlier periods until the DTLs reverse.
- Companies that incurred substantial losses in earlier years could apply tax credits (i.e., NOL carry-forwards) to reduce the amount of taxes due in later periods.
- When debt or accounts receivable is determined to be uncollectible (i.e., "Bad Debt" and "Bad AR"), this can create DTAs and tax differences. The expense can be reflected on the income statement as a write-off but not be deducted in the tax returns.

What are deferred tax liabilities (DTLs)?

Deferred tax liabilities ("DTLs") are created when a company recognizes a tax expense on its GAAP income statement that, because of a temporary timing difference between GAAP and IRS accounting, is not actually paid to the IRS that period but is expected to be paid in the future.

DTLs are often related to depreciation. Companies can use accelerated depreciation methods for tax purposes but elect to use straight-line depreciation for GAAP reporting. This means that for a given depreciable asset, the amount of depreciation recognized in the earlier years for tax purposes will be greater than under GAAP.

Those temporary timing differences are recognized as DTLs. Since these differences are just temporary – under both book and tax reporting, the same cumulative depreciation will be recognized over the life of the asset – at a certain point into the asset's useful life, an inflection point will be reached where the depreciation expense for tax reporting will become lower than for GAAP.

Temporary timing differences in book and tax accounting lead to the creation of DTAs/DTLs, which will gradually unwind to a balance of zero.

What are deferred tax assets (DTAs)?

Deferred tax assets ("DTAs") are created when a company recognizes a tax expense on its GAAP income statement that, due to a temporary timing difference between GAAP and IRS accounting rules, is lower than what must be paid to the IRS for that period. These net operating losses ("NOLs") that a company can carry forward against future income create DTAs.

For example, a company that reported a pre-tax loss of \$10 million will not get an immediate tax refund. Instead, it'll carry forward these losses and apply them against future profits.

However, under GAAP, the tax benefit will be recognized from a presumed future tax refund immediately on the income statement, and this difference gets captured in DTAs. As the company generates future profits and uses those NOLs to reduce future tax liabilities, the DTAs gradually reverse.

Another reason for DTAs is the differences between book and tax rules for revenue recognition. Broadly, tax rules require recognition based on receiving cash, while GAAP adheres rigidly to accrual concepts.

Read More → [Financial Concepts: Deferred Taxes](#)

What impact did the COVID-19 Tax Relief have on NOLs?

Under the Coronavirus Aid, Relief, and Economic Security (CARES) Act, NOLs that arise beginning in 2018 and through 2020 could be carried back for up to a maximum of five years. The rules for claiming tax losses were changed to assist individuals and corporations negatively impacted by the pandemic.

For tax years beginning after 2020, the CARES Act would allow NOLs deduction equal to the sum of:

1. All NOL carryovers from pre-2018 tax years
2. The lesser amount between 1) all NOL carryovers from post-2017 tax years or 2) 80% of remaining taxable income after deducting NOL carryovers from pre-2018 tax years

Previously, NOLs arising in tax years ending after 2017 could not be carried back to earlier tax years and offset taxable income. NOLs arising in tax years post-2017 could only be carried forward to later years. But the key benefit was that the NOLs could be carried forward indefinitely until the loss was fully recovered (yet limited to 80% of the taxable income in a single tax period).

What are the notable takeaways from Joe Biden's proposed tax plans?

- The corporate tax rate will rise from the Trump Era's Tax Cuts and Jobs Act ("TCJA") rate of 21% to 28%; estimated to increase the government's tax revenue from \$2 trillion to \$3 trillion over the next decade.
- The top tax rate for individuals with a taxable income of \$400k+ will rise from 37% to 39.6%.
- A 12.4% payroll tax will be imposed on those earning \$400k+ and to be split evenly between employers and employees.
- Minimum tax on corporations with book profits of \$100+ million, which would be structured so that corporations would pay the greater amount between 1) their regular corporate income tax or 2) the 15% minimum tax with net operating loss (NOL) and foreign tax credits allowed.

Does a company truly not incur any costs by paying employees through stock-based compensation rather than cash?

Stock-based compensation is a non-cash expense that reduces a company's taxable income and is added-back on the cash flow statement.

However, SBC incurs an actual cost to the issuer by creating additional shares for existing equity owners. The issuing company, due to the dilutive impact of the new shares, becomes less valuable on a per-share basis to existing shareholders.



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Could you define contra-liability, contra-asset, and contra-equity with examples of each?

- **Contra-Liability:** A contra-liability is a liability account that carries a debit balance. While classified as a liability, it functions closer to an asset by providing benefits to the company. An example would be financing fees in M&A. The financing fees are amortized over the debt's maturity, which reduces the annual tax burden and results in tax savings until the end of the term.
- **Contra-Asset:** A contra-asset is an asset that carries a credit balance. An example would be depreciation, as it reduces the fixed asset's carrying balance while providing tax benefits to the company. There is often a line called "Accumulated Depreciation," which is the contra-asset account reflected on the balance sheet.
- **Contra-Equity:** A contra-equity account has a debit balance and reduces the total amount of equity held by a company. An example would be treasury stock, which reduces shareholders' equity. Since treasury stock reduces the total shareholders' equity, treasury stock is shown as a negative on the balance sheet.

What is an allowance for doubtful accounts on the balance sheet?

Under US GAAP, the allowance for doubtful accounts estimates the percentage of uncollectible accounts receivable. This line item is considered a contra-asset because it reduces the accounts receivable balance. The allowance, often called a bad debt reserve, represents management's estimate of the amount of A/R that appears unlikely to be paid by customers. In effect, a more realistic value for A/R that'll actually be turning into cash is shown on the balance sheet, while preventing any sudden decreases in the company's A/R balance.

What is the difference between a write-down and a write-off?

- **Write-Downs:** In a write-down, an adjustment is made to an asset such as inventory or PP&E that has become impaired. The asset's fair market value (FMV) has fallen below its book value; hence, its classification as an impaired asset. Based on the write-down amount deemed appropriate, the value of the asset is decreased to reflect its true value on the balance sheet. Examples of asset write-downs would include damages caused by minor fires, accidents, or sudden value deterioration from lower demand.
- **Write-Offs:** Unlike a write-down in which the asset retains some value, a write-off reduces an asset's value to zero, meaning the asset has been determined to hold no current or future value (and should therefore be removed from the balance sheet). Examples include uncollectible AR, "bad debt," and stolen inventory.

How would a \$100 inventory write-down impact the three financial statements?

- **IS:** The \$100 write-down charge will be reflected in the cost of goods line item. The expense would decrease EBIT by \$100, and net income would decline by \$70, assuming a 30% tax rate.
 - **CFS:** The starting line item, net income, will be down \$70, but the \$100 write-down is an add-back since there's no actual cash outflow from the write-down. The net impact to the ending cash will be a \$30 increase.
 - **BS:** On the asset side, cash is up \$30 due to inventory being written down \$100. This will be offset by the decrease of \$70 in net income that flows through retained earnings on the equity section. Both sides of the balance sheet will be down by \$70 and remain in balance.
- The cash impact of the write-down was net positive, as the company's taxable income was reduced from the tax savings.**

How does buying a building impact the three financial statements?

- **IS:** Initially, there'll be no impact on the income statement since the purchase of the building is capitalized.
- **CFS:** The PP&E outflow is reflected in the cash from investing section and reduces the cash balance.
- **BS:** The cash balance will go down by the purchase price of the building, with the offsetting entry to the cash reduction being the increase in PP&E.

Throughout the purchased building's useful life, depreciation is recognized on the income statement, which reduces net income each year, net of the tax expense saved (since depreciation is tax-deductible).

How does selling a building with a book value of \$6 million for \$10 million impact the three financial statements?

- **IS:** If I sell a building for \$10 million with a book value of \$6 million, a \$4 million gain from the sale would be recognized on the income statement, which will increase my net income by \$4 million.
- **CFS:** Since the \$4 million gain is non-cash, it'll be subtracted from net income in the cash from operations section. In the investing section, the full cash proceeds of \$10 million are captured.
- **BS:** The \$6 million book value of the building is removed from assets while cash increases by \$10 million, for a net increase of \$4 million to assets. On the L&E side, retained earnings will increase by \$4 million from the net income increase, so the balance sheet remains balanced.

However, the gain on sale will result in higher taxes, which will be recognized on the income statement. This lowers retained earnings by \$1 million and be offset by a \$1 million credit to cash on the asset side.

If a company issues \$100 million in debt and uses \$50 million to purchase new PP&E, walk me through how the three statements are impacted in the initial year of the purchase and at the end of year 1. Assume a 5% annual interest rate on the debt, no principal paydown, straight-line depreciation with a useful life of five years and no residual value, and a 40% tax rate.

Initial Purchase Year (Year 0)

- **IS:** There'll be no changes as neither capex nor issuing debt impact the income statement.
- **CFS:** The \$50 million outflow of capex will be reflected in the cash from investing section of the cash flow statement, while the \$100 million inflow from the debt issuance will be reflected in the cash from financing section. The ending cash balance will be up by \$50 million.
- **BS:** On the assets side, cash will be up by \$50 million and PP&E will increase \$50 million from the PP&E purchase, making the assets side increase by \$100 million in total. On the L&E side, debt will be up \$100 million, which will offset the increase in assets and the balance sheet remains in balance.

End of First Year (Year 1)

- **IS:** Since the capex amount was \$50 million with a useful life assumption of five years (straight-line to a residual value of zero), the annual depreciation will be \$10 million. Next, the interest expense will be equal to the \$100 million in debt raised multiplied by the 5% annual interest rate, which comes out to \$5 million in annual interest expense. The pre-tax income will be down by \$15 million and assuming a 40% tax rate, net income will be down \$9 million.
- **CFS:** Net income will be down \$9 million, but the non-cash depreciation of \$10 million will be added back, making the ending cash balance increase by \$1 million.
- **BS:** On the assets side, cash is up by \$1 million and PP&E will decrease by \$10 million because of the depreciation. Since equity is also down \$9 million due to net income, both sides will remain in balance.

For long-term projects, what are the two methods for revenue recognition?

1. **Percentage of Completion Method:** In the percentage of completion method, revenue is recognized based on the percentage of work completed during the period. This method is used far more common since it's in a company's best interest to record partial revenue once earned. Two conditions must be met to use this method: the collection of payment must be reasonably assured, and the total project costs with the estimated completion date are required to be provided.
2. **Completed Contract Method:** The completed contract method recognizes revenue once the entire project has been completed. This method is rarely used in the US, as it would result in a company under-reporting revenue that has been earned under the accrual-based system.

If a company has continuously incurred goodwill impairment charges, what do you take away from seeing this in their financials?

Goodwill on the balance sheet remains unchanged unless it's impaired, meaning the purchaser has determined that the acquired assets are worth less than initially thought. While goodwill impairment can be attributed to unforeseeable circumstances, impairments as a common occurrence may raise concerns regarding the management team's history of overpaying for assets or their inability to integrate new acquisitions.

What is restricted cash and could you give me an example?

Restricted cash is cash reserved for a specific purpose and not available for the company to use. On the balance sheet, the restricted cash will be listed separated from the unrestricted cash, and there'll be an accompanying disclosure providing the reasoning why this cash cannot be used.

An example of restricted cash would be if a company signed an agreement to receive a line of credit, but the lender has required the borrower to maintain 10% of the total loan amount at all times (i.e., "bank loan requirement"). As long as the line of credit is active, the 10% minimum must be preserved to avoid breaching the lending terms. The company may have a separate bank account to hold the funds, or the lender may have required the amount to be placed in escrow to ensure compliance.

What is the accounting treatment for finance leases?

Finance leases is an accounting approach that recognizes the present value of all future lease payments as debt and recognizes the value provided by those future lease obligations as an asset (PP&E) on the lessee's balance sheet. Unlike debt, where the principal is defined, companies must estimate the initial finance lease liability as the present value of all future lease payments, using a discount rate assumption. For example, a 4-year lease with \$500,000 annual year-end lease payments at an assumed discount rate of 10% will be recognized.

Similar to debt, leases are long-term obligations to make payments to another party. However, lease payments rarely include explicit interest payments in the way debt does. Instead, the interest fees are implied and accounted for in the total lease expense.

Over the finance lease term, the asset is depreciated, while the lease liability accrues interest during the year and is then reduced by lease payments (similar to principal payments with debt). On the income statement, depreciation and the implied interest expense reduces net income.

To recap, the balance sheet initially treats the finance lease as a debt-like liability and the underlying asset as an owned asset. Over the life of the lease, the income statement impact doesn't capture the rent expense as one might intuitively assume. Instead, finance lease accounting breaks up the lease payments into two components on the income statement: interest and depreciation expenses – even though a company in actuality is paying a lease payment that commingles these two items.

What is the accounting treatment for operating leases?

Lease accounting changed significantly in 2019 for both US GAAP and IFRS. IFRS doesn't allow operating leases at all, so this only applies to US GAAP.

The initial balance sheet impact is the same as finance leases: Initially, the lease is recognized as a liability on the balance sheet (just like debt) with the corresponding asset as PP&E. The income statement is where the accounting diverges from finance leases. The income statement is simply reduced by the rent (lease) expense throughout the lease term. For example, if a 5-year lease calls for the annual lease payment of \$500,000, the annual rent expense will be recognized as a \$500,000 operating expense per year. The cash flow statement will already reflect the lease payment in each period via the net income line, so the lease payment affects the cash flow statement in cash from operations.

What are the three different types of intercompany investments?

1. **Investments in Securities:** The investment in securities method is used when a company invests in another company's equity, but the ownership percentage is less than 20%. These investments are treated as minor, passive financial investments due to the insignificant influence.
2. **Equity Investments Method:** When a company owns between 20-50% of another company, this is considered a significant level of influence. Thus, proper accounting treatment would be the equity method. Under the equity method, an investment is initially measured at the acquisition price and recorded as an "Investment in Affiliate" (or "Investment in Associate") on the assets side. Although a non-controlling stake, this type of ownership is considered influential enough to affect the target's decisions.
3. **Consolidation Method:** When the parent company has majority control over 50% ownership, the consolidation method is used. Instead of creating an individual investment asset, the target company's balance sheet is consolidated with the acquirer. To reflect that the acquirer owns less than 100% of the consolidated assets and liabilities, a new equity line titled "Non-Controlling Interests" (NCI) is created, which captures the value of equity in the consolidated business held by non-controlling (minority) interests (other third parties).

What are the three sub-classifications of investment securities?

1. **Trading Securities:** These are debt or equity investments intended to generate short-term profits. The purchase amount of the security will be recorded at its initial cost on the balance sheet and periodically marked-to-market until sold. Any unrealized gain/(loss) will be recorded on the income statement throughout the holding period until the realized gain/(loss) when sold.
2. **Available-for-Sale Securities ("AFS"):** These are debt or equity securities held for the long-term but sold before maturity. The investment amount will be recorded at the initial cost on the balance sheet, marked-to-market until the sale, and categorized as either current or non-current. A distinction from trading securities is how unrealized gains or losses are not be reflected on the income statement, but recorded as "Accumulated Other Comprehensive Income" on the balance sheet. Once sold, the realized gain/(loss) will be recognized on the income statement.

- 3. Held-to-Maturity Securities ("HTM"):** These are long-term investments in debt securities held until the end of their term. HTM applies only to debt instruments, typically government bonds, certificates of deposit (CDs), and investment-grade corporate bonds, as they have fixed payment schedules and maturity dates. HTM securities are low risk, low return holdings since there's a low risk of default, and the long-term holding horizon helps mitigate many risks. The investment's original cost is recorded on the balance sheet (reported at amortized cost). However, the value of the investment is not adjusted following changes in its FMV due to the HTM classification.

Could you name an example of an asset that's exempt from the cost principle rule?

Mark-to-market accounting would record the asset at its current fair market value and then adjust the value to reflect what an asset would sell for today. There are a few exceptions to the cost principle rule, with one of them being marketable securities, which are highly liquid and traded on stock exchanges.

What is trapped cash and what benefit does it provide to companies?

Trapped cash refers to the accumulation of cash overseas by multinational companies. While not illegal, companies keep this cash offshore to avoid certain repatriation taxes if brought back to the US.

For example, Apple held ~\$250 billion overseas at one point, mainly in Ireland, which has been known as a "tax-haven." While CEO Tim Cook had to defend Apple's tax practices to Congress, there was no sign of illegal wrongdoing, and Cook's counterargument was that Apple paid all required tax payments related to sales conducted in the US and shifted the narrative towards how Apple is a global company.

During the Trump administration, fixing this issue and incentivizing US companies to bring their operations back to their home country was a key objective. But even after the tax cuts went into effect, many corporations retained significant amounts of cash abroad – well short of the predicted \$4 trillion expected to be brought back. For multinational companies with operations in several countries, there's no clear incentive or obligation why they would have to bring all their cash and operations back to the US.

When can a company capitalize software development costs under accrual accounting?

The capitalization of software development costs involves internally developed software costs being recognized similar to fixed asset purchases, as opposed to being expensed as incurred in the current period. These software-related costs can include programmers' compensation, market testing, and various direct or indirect overhead costs related to bringing the software to the public. To capitalize software development costs, the software being developed must be eligible based on GAAP's criterion.

Broadly, there are two stages of software development in which a company can capitalize software costs:

1. The application development stage for software intended for internal use such as coding
2. The stage when the software's "technological feasibility" has been reached and can be marketed

The accounting treatment of capitalized software costs is like that of certain intangible assets, in which the software costs are capitalized and amortized over the useful life assumption on the income statement.

Read More → [Accounting for Capitalized Software Costs](#)

What is PIK interest?

PIK stands for "Paid-in-Kind." PIK interest expense is interest charged by a lender that accrues towards the ending debt balance as opposed to being paid in cash in the current period. While opting for PIK may conserve cash for the time being from deferring interest payments to a later date, the debt principal due at maturity increases each year, as well as the accrued interest payment amount.

What is a PIK toggle note?

A PIK toggle note provides the issuer of the debt with the option to defer an interest payment, but the entire debt balance, including the accrued payments that must be paid by maturity. Loans can come arranged with a fixed PIK schedule outlined in the lending terms, but certain debt instruments can come with the optionality to let the issuer decide whether to pay in cash or accrue over to the next period. Therefore, whether interest expense is paid-in-cash or PIK effectively becomes a discretionary decision.

If a company has incurred \$100 in PIK interest, how would the three-statements be impacted?

- **IS:** On the income statement, interest expense will increase by \$100, which causes EBIT to decrease by \$100. Assuming a 30% tax rate, net income will decrease by \$70.
- **CFS:** On the cash flow statement, net income will be down by \$70, but the \$100 non-cash PIK interest will be added back. The ending cash balance will increase by \$30.
- **BS:** On the assets side of the balance sheet, cash will be up \$30. Then on the liabilities side, the debt balance will be up \$100 (since the PIK accrues to the debt's ending balance), and net income is down \$70. Therefore, both sides are up by \$30, and the balance sheet balances.

What is the purpose of the original issue discount feature of debt?

An original issue discount ("OID") is a feature of debt meant to make the issuance more appealing to investors as a "deal sweetener." Typically, the reduction is 1-2% of the debt issuance.

For example, that if there's \$100 of debt being raised, you'll issue it so that the lender only has to pay \$98 or \$99. An OID is modeled similarly to financing fees and amortized over the term of the debt.

Why are circularities created in financial models?

A circularity is introduced into a financial model when a cell, either directly or indirectly, refers to itself. The most common circularity is created by interest expense and interest income.

For example, interest expense is calculated off the beginning and ending balances of a company's debt outstanding, which includes the revolver. Furthermore, the revolver drawdown/(paydown) for a given period is directly impacted by interest expense – thereby, a circular reference is created.

How would you forecast a company's basic and diluted share count?

In general, management teams will publicly announce their plans for future share issuances or a buyback program. If that's the case, then the number of share issuance and repurchases can be calculated using the formula shown below:

$$\text{Basic Share Count EOP} = \text{Basic Share Count BOP} + \left(\frac{\text{Shares Issued}}{\text{Estimated Share Price}} \right) - \left(\frac{\text{Share Repurchases}}{\text{Estimated Share Price}} \right)$$

The estimated share price used above can be approximated using the formula below:

$$\text{Share Price Estimate} = \text{Prior Period Share Price} \times (1 + \text{Current Period Consensus EPS Growth Rate})$$

To calculate the impact of dilutive securities, the differences between the basic and diluted share count in historical periods can be looked at and this amount can be straight-lined into the future. While this approach has its shortcomings, it serves as a decent proxy for how potentially dilutive securities such as options, warrants, and convertible debt will impact the diluted share count.

How can you forecast a company's implied share price using its EPS?

Since the P/E ratio is equal to a company's share price divided by its EPS, a company's implied share price can be estimated by taking the forecasted EPS figure and then multiplying it by a P/E ratio assumption. The conservative approach is to use the company's current P/E ratio as of the present day (or a minor contraction).

This estimated share price can be sanity checked by using the consensus EPS annual growth rate as a proxy for share price growth. Both approaches implicitly assume a fixed P/E ratio, but this is meant to be an approximation of where a company could be trading at, rather than a precise share price forecast.

Read More → [Forecasting a Company's Shares Outstanding and Earnings Per Share](#)

What are the two types of pension plans and how does the accounting differ for each?

A pension plan is a contract between an employer and employee, in which the employer agrees to pay cash benefits to the employee upon retirement.

1. **Defined Contribution:** The plan is the simpler of the two, as the employer makes specified contributions into the plan periodically. The accounting treatment of defined contribution plans is fairly straightforward as a pension expense will be shown on the income statement (typically SG&A).
2. **Defined Benefit:** The other option involves the employer having to estimate each period how much of a contribution must be made to satisfy the upcoming defined post-retirement benefits (and thus, there will be assumptions on expected payments required). For defined benefit plans, the accounting becomes more complicated since it involves expected values. But the defined benefit plan will recognize a pension expense in the SG&A with the off-setting balance sheet entry being either a liability if the amount of cash contribution is smaller than the pension expense recorded on the income statement or an asset as a pre-paid expense if the amount of cash contribution was greater than the pension expense recorded on the income statement. Both would lead to tax implications since there would be differences between GAAP and IRS taxes, which creates DTAs/DTLs.